

E0 270(0): Assignment 2

Due date: March 28, 2026

Instructions

1. You are allowed to change the template code according to your requirements.
2. Setting up the environment, running into technical problems, and figuring out their solutions is a part of the learning process. Use Google, Stack overflow, and discuss with each other. Unfortunately, we will not be able to help you.
3. The report must be typeset in $\text{\LaTeX}$, Ensure that everything is in a single pdf file and all pages are in correct order.
4. If you feel any particular problem is under-specified, you can safely assume that it has been done intentionally. You are free to make assumptions, but please specify the assumptions in your report. No further clarifications on the problem will be provided. However feel free to contact us if you believe that there is an error in the problem specification.
5. The grading will be relative based on the final performance achieved by your model. You are encouraged to discuss the problems amongst yourselves and engage in a healthy competition. However copying codes from others verbatim, along with the use of LLM assisted code generators are strictly forbidden, and will be penalized heavily if detected.

Data

Consider the Breast Cancer Wisconsin (Diagnostic) dataset(data.csv), where the label is "Diagnosis" with two classes: M = malignant, B = benign.

Logistic Regression

Logistic Regression is a discriminative model that predicts the probability of a binary outcome by passing a linear combination of features through a logistic (sigmoid) function. You are required to implement Logistic Regression on the provided data with the following instructions.

1. Preprocessing: Use the same 70/30 stratified split as train/ test. You must standardize (z-score normalize) the features using the training set and apply the same transformation to the validation and test sets.
2. Model: Implement Logistic Regression using Gradient Descent from scratch (do not use scikit-learn's LogisticRegression). Use Binary Cross-Entropy (BCE) as your loss function. Train your model using different learning rates $\alpha \in \{0.1, 0.01, 0.001\}$ and plot the Loss vs. Iterations curve for each to observe convergence.
3. Evaluation: Report accuracy, precision, recall, F1, and ROC-AUC on the test set. Extract the learned weights, identify the top 3 features with the largest absolute weights, and briefly explain what this implies about feature importance. Finally, pick the top 2 features and plot the linear decision boundary alongside a scatter plot of those features.

Multi-Layer Perceptron (MLP)

A Multi-Layer Perceptron is a feedforward artificial neural network capable of modeling complex, non-linear relationships.

1. Preprocessing: Utilize the same standardized dataset prepared in the Logistic Regression section.
2. Model: Experiment with three different architectures: a linear baseline (0 hidden layers), a wide network (1 hidden layer, 100 neurons), and a deep network (3 hidden layers, 10 neurons each). For the deep network, compare the convergence speed when using Sigmoid versus ReLU activation functions in the hidden layers.
3. Evaluation: Plot Training Accuracy vs. Validation Accuracy over the training epochs for your best model to analyze potential overfitting. Report the final test metrics (accuracy, precision, recall, F1, ROC-AUC). Finally, provide a brief comparative analysis: does the MLP significantly outperform Logistic Regression on this dataset, and how does the trade-off between performance and interpretability apply here?

Deliverables

- Completed code for implementing the Logistic Regression and MLP in the given incomplete code snippets.
- A report containing the complete details of the algorithm and your implementation, along with justification for any extra assumptions made if necessary.
- Training plot corresponding to accuracies for both train and test set.

Submission

Attach a **single zip file** named in the format `Asst1_FirstName_LastName_5DigitsOfSRNo.zip` to the assignment in Teams, before the due date. The zip file should contain your code that you filled up, your report (in a pdf file named `Report_<5digitSR>.pdf`), and any plots that you put in the report. Ensure to upload the zip file on Teams submission portal on or before March 9, 2026, 11:59 pm. Ensure that you do **not** include the data files and **pycache** files in the code that you submit.